ZIYUE YU

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EDUCATION

University of Pittsburgh *September 2024 – Present*
Ph.D. in Civil and Environmental Engineering
Advisor: Prof. Lei Fang

Southern University of Science and Technology *September 2020 – June 2024*
B.Eng. in Mechanical and Aerospace Engineering
Advisor: Prof. Hongyan Yuan

RESEARCH INTERESTS

Low-Reynolds-number turbulence, scale-to-scale energy and enstrophy transfer in quasi-two-dimensional flows, Lagrangian coherent structures, and active-matter–flow interactions, with a focus on regimes where inertia is weak but multiscale transport persists.

PROGRAMS AND JOURNAL PUBLICATIONS (ADVISOR UNDERLINED)

<Tensor-Geometry-Activated Spectral Energy Transfer at Reynolds Numbers of Order Unity>

Z. Yu, X. Si, and L. Fang, "Tensor-geometry-activated spectral energy transfer at Reynolds numbers of order unity in two-dimensional flows," *Journal of Fluid Mechanics* **1044**, A32 (2026).
Aligning the stress and strain-rate tensors with small directional perturbations amplifies the scale-to-scale energy flux up to $\sim$300$\times$ at $Re \sim O(1)$, crossing the weak-turbulence threshold in a nominally laminar flow.

<Two-Way Coupling in Active Suspensions>

Z. Yu, M. Wu, and L. Fang, "Two-way coupling in active suspensions suppresses particle accumulation and induces non-monotonic flow stabilization," manuscript being finalized for *Physical Review Fluids*.
In 85 experiments with live swimmers in an electromagnetically driven cellular flow, the swimmers' back-reaction suppresses LCS accumulation and can either disrupt or reinforce the flow's transport skeleton, giving a non-monotonic stability phase diagram.

<Filter-Space Decomposition of the Enstrophy Flux in Two-Dimensional Turbulence>

Z. Yu and L. Fang, "Filter-space decomposition of the enstrophy flux in two-dimensional turbulence: Leonard artifacts and cross-term transfer," manuscript in preparation.
Extends the Leonard/Cross/SGS decomposition to the enstrophy flux: the Leonard term is a measurement artifact, while the forward enstrophy cascade lives in the cross term. Validated with 1024^2–4096^2 DNS and experiments at $Re \approx 11$–959.

<Three-Dimensional Active Suspensions: Alignment-Selected Inverse Energy Transfer>

Z. Yu and L. Fang, research in progress.
Combines multi-camera 3D Lagrangian tracking (Shake-the-Box) of swimming brine shrimp with OpenFOAM simulations of jet arrays to test whether collective small-scale activity drives an inverse energy cascade by enlarging tensor-aligned shear regions.

CONFERENCE PRESENTATIONS

Z. Yu and L. Fang, "Low Reynolds number turbulence: Tensor geometry drives turbulence in the diminishing of inertia," 78th Annual Meeting of the APS Division of Fluid Dynamics, Houston, TX, November 2025.

RESEARCH METHODS

Experimental: 3D Particle Tracking Velocimetry (PTV) and multi-camera Shake-the-Box Lagrangian tracking; electromagnetically forced thin-layer flows; Arduino-based circuit design and motor control; active-matter cultivation.

Numerical Simulation: OpenFOAM simulations; Ansys Fluid; Abaqus.

Data Analysis: MATLAB (advanced), Python; energy/enstrophy flux, spectral, and FTLE/LCS analysis; image and flow-field processing.

Tools: Claude Code (AI-assisted coding and research workflows); Git; LaTeX.

TEACHING EXPERIENCE

Teaching Assistant, University of Pittsburgh *2024–2025*
CEE 1402: Fluid Mechanics
CEE 1610: Engineering and Sustainable Development

AWARDS AND HONORS

Teaching Assistant of the Year, Department of Civil and Environmental Engineering, Pitt *2024–2025*

Second-Class Scholarship for Outstanding Students, SUSTech *2022*

Fastest Progress Scholarship, Department of Mechanical and Aerospace Engineering, SUSTech *2022*

Star of Literature and Art, SUSTech *2022*

Outstanding Student Advisor, SUSTech *2022*

Third-Class Scholarship for Outstanding New Undergraduates, SUSTech *2020*